Student Number:									

3 Complex Analysis 2023

A/Prof Elena Berdysheva
Mita Ramabulana

Test 2

11 October 2023

Instructions:

- This question paper consists of **6 pages** (including this one).
- You have 90 minutes to answer the questions.
- Answer all questions in the spaces provided on this question paper. Use the backs of pages if needed.
- Use your *UCT Examination Answer Book* for rough work. The work in your UCT Examination Answer Book will not be marked. Only the answers that you write on this question paper will be marked.
- Give full proofs.
- Be careful to provide answers that we can read and make sense of at all times. If we can't read your handwriting, we will mark your solution incorrect.
- 40 points = 100% (42 points available).

Problem	1	2	3	4	5	Σ	Mark
Maximum Points	8	10	8	8	8	40	100%
Result							

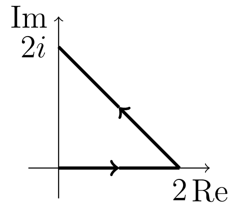
Student Number:									

Problem 1

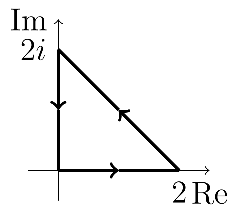
(8 points)

Calculate the following integrals.

(a) $\int_\gamma z^2\,dz$, where γ is the indicated path:



(b) $\int_\gamma z^2\,dz$, where γ is the indicated path:



Solution.
Notice that the function $f(z) = z^2$ has an antiderivative $F(z) = \frac{1}{3}z^3$ on $\mathbb{C}$. We are going to use the Fundamental Theorem of Calculus.

(a) $\displaystyle\int_\gamma z^2\,dz = F(2i) - F(0) = \frac{1}{3}(2i)^3 - 0 = -\frac{8i}{3}.$

(b) In this case the path is closed, and therefore $\displaystyle\int_\gamma z^2\,dz = 0.$

Student Number:									

Problem 2

(10 points)

Use Cauchy's Integral Formula and Cauchy's Integral Formula for derivatives to calculate the following integrals.

(a) $\displaystyle\int_\gamma \frac{\cos \pi z}{z^2-1}\,dz$, where $\gamma(t) = 3e^{it}$, $t \in [0, 2\pi]$.

(b) $\displaystyle\int_\gamma \frac{e^z}{(z-2)^2}\,dz$, where $\gamma(t) = 2 + e^{it}$, $t \in [0, 2\pi]$.

Solution.

(a) We first determine the partial fraction decomposition of $\frac{1}{z^2-1}$:

$$\frac{1}{z^2-1} = \frac{A}{z-1} + \frac{B}{z+1} = \frac{(A+B)z + (A-B)}{z^2-1},$$

and we must have $A + B = 0$, $A - B = 1$. The solution is $A = \frac{1}{2}$, $B = -\frac{1}{2}$. Thus,

$$\frac{1}{z^2-1} = \frac{1}{2}\frac{1}{z-1} - \frac{1}{2}\frac{1}{z+1}.$$

For the integral we have

$$\begin{aligned}\int_\gamma \frac{\cos \pi z}{z^2-1}\,dz &= \frac{1}{2}\int_\gamma \frac{\cos \pi z}{z-1}\,dz - \frac{1}{2}\int_\gamma \frac{\cos \pi z}{z+1}\,dz \\ &= \pi i \cos \pi - \pi i \cos(-\pi) = 0.\end{aligned}$$

(b) We use Cauchy's Integral Formula for derivative with $n = 1$:

$$\int_\gamma \frac{e^z}{(z-2)^2}\,dz = 2\pi i\ (e^z)'|_{z=2} = 2\pi i e^2.$$

Student Number:									

Problem 3

(8 points)

(a) Formulate the Coincidence Principle for functions f and g that are holomorphic in a region $G \subseteq \mathbb{C}$.

(b) Does there exists a function f that is holomorphic in the open disc $D(0;3) = \{z \in \mathbb{C} : |z| < 3\}$ such that

$$f\left(\frac{n-1}{n}\right) = f\left(\frac{n+1}{n}\right) = \frac{1}{n} \quad \text{for all } n \in \mathbb{N}?$$

Justify your answer.

Solution.

(a) Let f and g be holomorphic in a region $\mathbb{C}$. If the set $\{z \in G : f(z) = g(z)\}$ has a limit point in G, then $f \equiv g$.

(b) Both sets $\left\{\frac{n-1}{n} : n \in \mathbb{N}\right\}$ and $\left\{\frac{n+1}{n} : n \in \mathbb{N}\right\}$ have a limit point $1 \in D(0;3)$.

If $f\left(\frac{n-1}{n}\right) = f\left(1 - \frac{1}{n}\right) = \frac{1}{n}$, then by the Coincidence Principle f coincides in $D(0;3)$ with the function with the property $f(1-z) = z$, i.e. $f(z) = 1 - z$.

If $f\left(\frac{n+1}{n}\right) = f\left(1 + \frac{1}{n}\right) = \frac{1}{n}$, then by the Coincidence Principle f coincides in $D(0;3)$ with the function with the property $f(1+z) = z$, i.e. $f(z) = z - 1$.

But this is impossible at the same time. We conclude that there is no such function f.

Student Number:									

Problem 4

(8 points)

Show that there exists no holomorphic function $f : \mathbb{C} \to \mathbb{C}$ with

$$|f'(0)| > \max_{|z|=1} |f(z)|.$$

Solution.
Consider the unit circle $\gamma(t) = e^{it}$, $t \in [0, 2\pi]$. By Cauchy's Integral Theorem for derivatives,

$$f'(0) = \frac{1}{2\pi i} \int_\gamma \frac{f(z)}{(z-0)^2}\, dz.$$

We are going to use the Estimation Theorem. If z is on the unit circle, then $|z|^2 = 1$. The length of the unit circle is 2π. We obtain

$$|f'(0)| = \left| \frac{1}{2\pi i} \int_\gamma \frac{f(z)}{(z-0)^2}\, dz \right| \leq \frac{1}{2\pi} \max_{|z|=1} |f(z)| \cdot 2\pi = \max_{|z|=1} |f(z)|.$$

Therefore there is no holomorphic function $f : \mathbb{C} \to \mathbb{C}$ with

$$|f'(0)| > \max_{|z|=1} |f(z)|.$$

Student Number:									

Problem 5

(8 points)

Find all singularities of the function $f(z) = \dfrac{(z+1)(z-i)}{(z^2+1)(z+i)}$ and determine their type (a removable singularity, a pole, or an essential singularity). For a removable singularity, give a holomorphic extension of f. For a pole, give its order.

Solution.
Notice that $(z^2+1)(z+i) = (z-i)(z+i)(z+i) = (z-i)(z+i)^2$, i.e.

$$f(z) = \frac{(z+1)(z-i)}{(z-i)(z+i)^2}.$$

The singularities are $z_1 = i$ and $z_2 = -i$.
The point $z_1 = i$ is a removable singularity. Indeed, there exists the limit

$$\lim_{z\to i} f(z) = \lim_{z\to i} \frac{(z+1)(z-i)}{(z-i)(z+i)^2} = \lim_{z\to i} \frac{(z+1)}{(z+i)^2} = \frac{i+1}{(2i)^2} = -\frac{1}{4}(1+i).$$

The holomorphic extension is $\tilde{f}(z) = \frac{(z+1)}{(z+i)^2}$, $z \in \mathbb{C} \setminus \{-i\}$.
The point $z_2 = -i$ is a pole of degree 2. Indeed,

$$f(z) = \frac{\varphi(z)}{(z+i)^2},$$

where $\varphi(z) = \frac{(z+1)(z-i)}{(z-i)}$ is holomorphic in a disc $D(-i; r)$ with a small r (for example, we can take $r = 1$).